

Machine Learning

Homework 2

Nov. 2025

Fill your Name
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1. For the support vector machine (SVM), the data is not nearly separable in the real world application, so we need to modify SVM by introducing an error margin that must be minimized.

Specifically, the formulation we have looked at is known as the ℓ_1 norm soft margin SVM. In this problem we will consider an alternative method, known as the ℓ_2 norm soft margin SVM. This new algorithm is given by the following optimization problem (notice that the slack penalties are now squared):

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + \frac{C}{2} \sum_{i=1}^N \xi_i^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)} + b) \geq 1 - \xi_i, i = 1, \dots, N \end{aligned} \tag{1}$$

- (a) Notice that we have dropped the $\xi_i \geq 0$ constraint in the ℓ_2 problem. Show that these non-negativity constraints can be removed. That is, show that the optimal value of the objective will be the same whether or not these constraints are present.
- (b) What is the Lagrangian of the ℓ_2 soft margin SVM optimization problem?
- (c) Minimize the Lagrangian with respect to w , b , and ξ by taking the following gradients: $\nabla_w \mathcal{L}$, $\frac{\partial \mathcal{L}}{\partial b}$ and $\nabla_\xi \mathcal{L}$, and then setting them equal to 0. Here, $\xi = [\xi_1, \dots, \xi_N]^T$.
- (d) What is the dual of the ℓ_2 soft margin SVM optimization problem?

Ans:

- (a)
- (b)
- (c)
- (d)

2. Consider the following convex optimization problem:

$$\begin{aligned} \min_{x_1, \dots, x_n} \quad & - \sum_{i=1}^n \log(\alpha_i + x_i) + \sum_{i=1}^n \beta_i x_i \\ \text{s.t.} \quad & x_i \geq 0, \sum_{i=1}^n x_i = 1, \end{aligned} \tag{2}$$

where $\alpha_i > 0$ and β is a parameter.

- (a) Write the KKT conditions for the problem.
- (b) Find the optimal solution of the problem.

Ans:

- (a)
- (b)

3. Given four binary classification datasets, SVMs with different kernel functions were used for classification, and the following decision boundaries were obtained. Please analyze the characteristics of each decision boundary, determine the type of kernel function used, and explain your reasoning.

- (a) A straight line perfectly separates two types of data.
- (b) Smooth curved boundaries, presenting as circles or ellipses.
- (c) The polygonal boundary has distinct corners.

Ans:

- (a)
- (b)
- (c)